

Experiment 9

Aim: To understand Docker Architecture and Container Life Cycle, install Docker and execute docker commands to manage images and interact with containers.

Theory:

Docker is a popular platform that enables developers to build, package, and deploy applications as lightweight, portable, and self-sufficient containers. These containers encapsulate all the necessary dependencies and libraries required for an application to run, ensuring consistency across different environments. Here is a theoretical overview of

Docker:

Containerization:

Docker utilizes containerization technology to create isolated environments for applications. Containers are lightweight, standalone, and executable packages that include everything needed to run an application, such as code, runtime, system tools, libraries, and settings. This isolation ensures that applications run consistently across different environments, from development to production.

Docker Engine:

At the core of Docker is the Docker Engine, which is responsible for building, running, and managing containers. It consists of the Docker daemon, which manages containers, images, networks, and volumes, and the Docker client, which allows users to interact with the daemon through the Docker API.

Docker Images:

Docker images are read-only templates used to create containers. They contain the application code, runtime, libraries, dependencies, and other files needed to run the application. Images are built using Dockerfiles, which are text files that define the steps needed to create the image.

Docker Containers:

Containers are instances of Docker images that are running as isolated processes on a host machine. They are lightweight, portable, and can be easily started, stopped, moved, and deleted. Containers provide a consistent environment for applications to run, regardless of the underlying infrastructure.

Benefits of Docker:

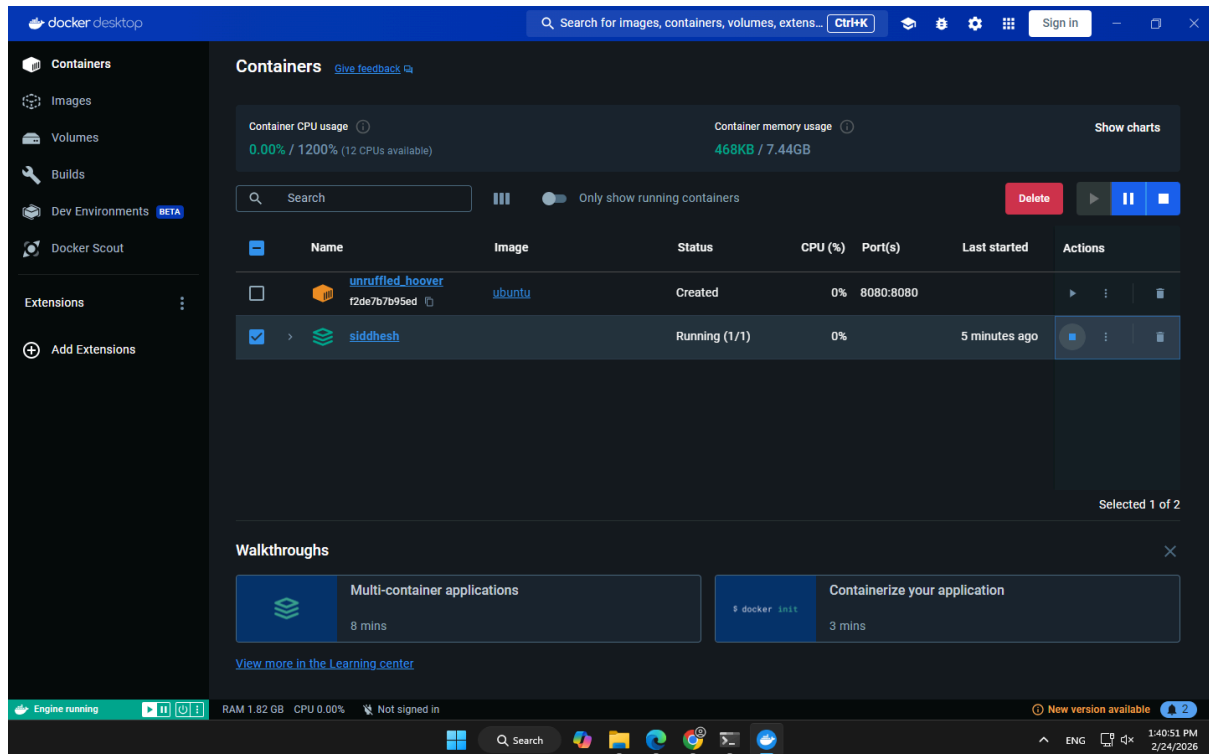
Portability: Docker containers can run on any platform that supports Docker, making it easy to deploy applications across different environments.

Efficiency: Containers share the host OS kernel, reducing overhead and improving resource utilization.

Isolation: Containers provide a level of isolation that helps prevent conflicts between applications and dependencies.

Scalability: Docker enables easy scaling of applications by quickly spinning up additional containers.

Consistency: Docker ensures that applications run the same way in development, testing, and production environments.



```
C:\Users\Lenovo>docker --help

Usage:  docker [OPTIONS] COMMAND

A self-sufficient runtime for containers

Common Commands:
  run         Create and run a new container from an image
  exec        Execute a command in a running container
  ps          List containers
  build        Build an image from a Dockerfile
  pull        Download an image from a registry
  push        Upload an image to a registry
  images      List images
  login        Log in to a registry
  logout       Log out from a registry
  search       Search Docker Hub for images
  version      Show the Docker version information
  info         Display system-wide information

Management Commands:
  builder      Manage builds
  buildx*      Docker Buildx (Docker Inc., v0.12.1-desktop.4)
  compose*     Docker Compose (Docker Inc., v2.24.6-desktop.1)
  container    Manage containers
  context       Manage contexts
  debug*       Get a shell into any image or container. (Docker Inc., 0.0.24)
  dev*         Docker Dev Environments (Docker Inc., v0.1.0)
  extension*   Manages Docker extensions (Docker Inc., v0.2.22)
  feedback*    Provide feedback, right in your terminal! (Docker Inc., v1.0.4)
  image        Manage images
  init*        Creates Docker-related starter files for your project (Docker Inc., v1.0.1)
  manifest     Manage Docker image manifests and manifest lists
  network      Manage networks
  plugin       Manage plugins
  sbom*        View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc., 0.6.0)
  scout*       Docker Scout (Docker Inc., v1.5.0)
  system       Manage Docker
  trust        Manage trust on Docker images
  volume       Manage volumes
```

```
C:\Users\Lenovo>docker --version
Docker version 25.0.3, build 4deb41
```

```
C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker search ubuntu

NAME                DESCRIPTION                                STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys... 17789     [OK]
ubuntu/squid         Squid is a caching proxy for the Web. Long-t... 124
ubuntu/nginx         Nginx, a high-performance reverse proxy & we... 139
ubuntu/cortex        Cortex provides storage for Prometheus. Long... 4
ubuntu/bind9         BIND 9 is a very flexible, full-featured DNS... 117
ubuntu/kafka         Apache Kafka, a distributed event streaming ... 60
ubuntu/zookeeper     ZooKeeper maintains configuration informatio... 14
ubuntu/apache2       Apache, a secure & extensible open-source HT... 102
ubuntu/prometheus    Prometheus is a systems and service monitori... 78
ubuntu/mysql         MySQL open source fast, stable, multi-thread... 72
ubuntu/dotnet-aspnet  Chiselled Ubuntu runtime image for ASP.NET a... 26
ubuntu/jre           Chiselled Java runtime based on Ubuntu. Long... 24
ubuntu/postgres      PostgreSQL is an open source object-relatio... 42
ubuntu/python        A chiselled Ubuntu rock with the Python runt... 31
ubuntu/redis         Redis, an open source key-value store. Long... 24
ubuntu/dotnet-runtime Chiselled Ubuntu runtime image for .NET apps... 24
ubuntu/dotnet-deps   Chiselled Ubuntu for self-contained .NET & A... 16
ubuntu/grafana        Grafana, a feature rich metrics dashboard & ... 12
ubuntu/memcached     Memcached, in-memory keyvalue store for smal... 6
ubuntu/prometheus-alertmanager Alertmanager handles client alerts from Prom... 10
ubuntu/cassandra     Cassandra, an open source NoSQL distributed ... 3
ubuntu/mlflow        MLFlow: for managing the machine learning li... 6
ubuntu/loki          Grafana Loki, a log aggregation system like ... 2
ubuntu/mimir        Ubuntu ROCK for Mimir, a horizontally scalab... 0
ubuntu/opentelemetry-collector This is a rock for the OpenTelemetry Collect... 0
```

```
C:\Users\Lenovo>docker search --filter=stars=1000 ubuntu

NAME                DESCRIPTION                                STARS     OFFICIAL
ubuntu              Ubuntu is a Debian-based Linux operating sys... 17789     [OK]
```

```
C:\Users\Lenovo>docker search mysql
```

NAME	DESCRIPTION	STARS	OFFICIAL
mysql	MySQL is a widely used, open-source relation...	16070	[OK]
bitnami/mysql	Bitnami Secure Image for mysql	147	
circledi/mysql	MySQL is a widely used, open-source relation...	32	
bitnamicharts/mysql	Bitnami Helm chart for MySQL	0	
cimg/mysql		3	
ubuntu/mysql	MySQL open source fast, stable, multi-thread...	72	
google/mysql	MySQL server for Google Compute Engine	26	
linuxserver/mysql	A Mysql container, brought to you by LinuxSe...	48	
elestio/mysql	Mysql, verified and packaged by Elestio	3	
alpine/mysql	mysql client	7	
docksal/mysql	MySQL service images for Docksal - https://d...	0	
eclipse/mysql	Mysql 5.7, curl, rsync	3	
ilios/mysql	Mysql configured for running Ilios	1	
datajoint/mysql	MySQL image pre-configured to work smoothly ...	2	
ddev/mysql	ARM64 base images for ddev-database-mysql-8...	2	
mirantis/mysql		0	
mysql/mysql-server	Optimized MySQL Server Docker images. Create...	1033	
corpusops/mysql	https://github.com/corpusops/docker-images/	0	
cleanstart/mysql	Secure by Design, Built for Speed, Hardened ...	0	
vulhub/mysql		1	
cbioportal/mysql	This repository hosts MySQL database images ...	1	
vitess/mysql	Lightweight image to run MySQL with Vitess	1	
mysql/mysql-router	MySQL Router provides transparent routing be...	28	
mysql/mysql-cluster	Experimental MySQL Cluster Docker images. Cr...	100	
nasqueron/mysql		1	

```
C:\Users\Lenovo>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
Digest: sha256:d1e2e92c075e5ca139d51a140ffff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

What's Next?

1. Sign in to your Docker account → [docker login](#)
2. View a summary of image vulnerabilities and recommendations → [docker scout quickview ubuntu](#)

```
C:\Users\Lenovo>
```

```
C:\Users\Lenovo>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
ubuntu	latest	bdbabce66f1b	13 days ago	78.1MB
ubuntu	<none>	802541663949	6 months ago	78.1MB
ubuntu	<none>	ca2b0f26964c	24 months ago	77.9MB
redis	<none>	170ale90f843	2 years ago	138MB
docker/dev-environments-default	stable-1	7eb85f44c229	3 years ago	610MB
crazymax/linguist	7.20.0	dc5f9ae3be6a	3 years ago	72.2MB
docker/desktop-git-helper	5a4fca126aadcd3f6cc3a011aa991de982ae7000	efe2d67c403b	4 years ago	44.2MB

```
C:\Users\Lenovo>docker run -it ubuntu
root@715e3a4a7ed8:/# exit
exit

C:\Users\Lenovo>docker run -it ubuntu
root@2ddd2c7718ec:/# admin
bash: admin: command not found
root@2ddd2c7718ec:/# ^C
root@2ddd2c7718ec:/# exit
exit

C:\Users\Lenovo>docker run -d httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
0c8d55a45c0d: Pull complete
fe1db8dd446f: Pull complete
4f4fb700ef54: Pull complete
30973b009bab: Pull complete
85b0d820aa56: Pull complete
996228ca33ab: Pull complete
Digest: sha256:b89c19a390514d6767e8c62f29375d0577190be448f63b24f511d6b03f7bf18
Status: Downloaded newer image for httpd:latest
9f106402c64c6c3743dc735fe26caf4a43a9876c261d5b29b20a340b446922393d
```

```
C:\Users\Lenovo>docker run -d mysql
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
74a8e4bbd9fe: Pull complete
35792636d31: Pull complete
f3702ac323ca: Pull complete
658630c937c6: Pull complete
ab1b8ea72826: Pull complete
61941d436e88: Pull complete
c89a19cdad3b: Pull complete
00af5f73db53: Pull complete
8314ff001dec: Pull complete
4a629f1008ff: Pull complete
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52659a
Status: Downloaded newer image for mysql:latest
af5bbf473e08d0b9a7264168926d9712c94a6d94497687fed89179d31e778ccd

C:\Users\Lenovo>docker run -d ubuntu
17ae4f0c3240c5b5ba81df3d23be8a1eef290d48684f89b47335567685514b7a

C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
9f10642c64c6   httpd      "httpd-foreground"      About a minute ago    Up About a minute    80/tcp       wonderful_babbage
a398ed3e2ae7   docker/dev-environments-default:stable-1   "sleep infinity"        23 months ago        Up 12 minutes                siddhesh-app-1
```